

EXPERIMENT-11

IMPLEMENTATION OF DIFFIE-HELLMAN KEY EXCHANGE

AIM

To implement the Diffie–Hellman key exchange algorithm for securely generating a common secret key between two users.

ALGORITHM

1. Choose a prime number p .
2. Choose a primitive root g .
3. User A chooses a private key a .
4. User B chooses a private key b .
5. Calculate A's public key: $A = g^a \bmod p$.
6. Calculate B's public key: $B = g^b \bmod p$.
7. Calculate the shared secret key:
 - User A: $K = B^a \bmod p$
 - User B: $K = A^b \bmod p$
8. Verify that both users obtain the same shared key.

CODE:

```
main.c
1 #include <stdio.h>
2 int power(int a, int b, int p)
3 {
4     int r = 1;
5     while(b-->0)
6         r = (r * a) % p;
7     return r;
8 }
9 int main()
10 {
11     int p, g, a, b, A, B, k1, k2;
12     printf("Enter prime number: ");
13     scanf("%d", &p);
14     printf("Enter primitive root: ");
15     scanf("%d", &g);
16     printf("Enter private key of A: ");
17     scanf("%d", &a);
18     printf("Enter private key of B: ");
19     scanf("%d", &b);
20     A = power(g, a, p);
21     B = power(g, b, p);
22     k1 = power(B, a, p);
23     k2 = power(A, b, p);
24     printf("\nPublic key of A: %d", A);
25     printf("\nPublic key of B: %d", B);
26     printf("\nShared key of A: %d", k1);
27     printf("\nShared key of B: %d", k2);
28     return 0;
}
```

OUTPUT:

```
Output
Enter prime number: 23
Enter primitive root: 5
Enter private key of A: 6
Enter private key of B: 15

Public key of A: 8
Public key of B: 19
Shared key of A: 2
Shared key of B: 2

=== Code Execution Successful ===
```

RESULT:

Thus, the Diffie–Hellman key exchange algorithm was successfully implemented, and both users generated the same shared secret key.